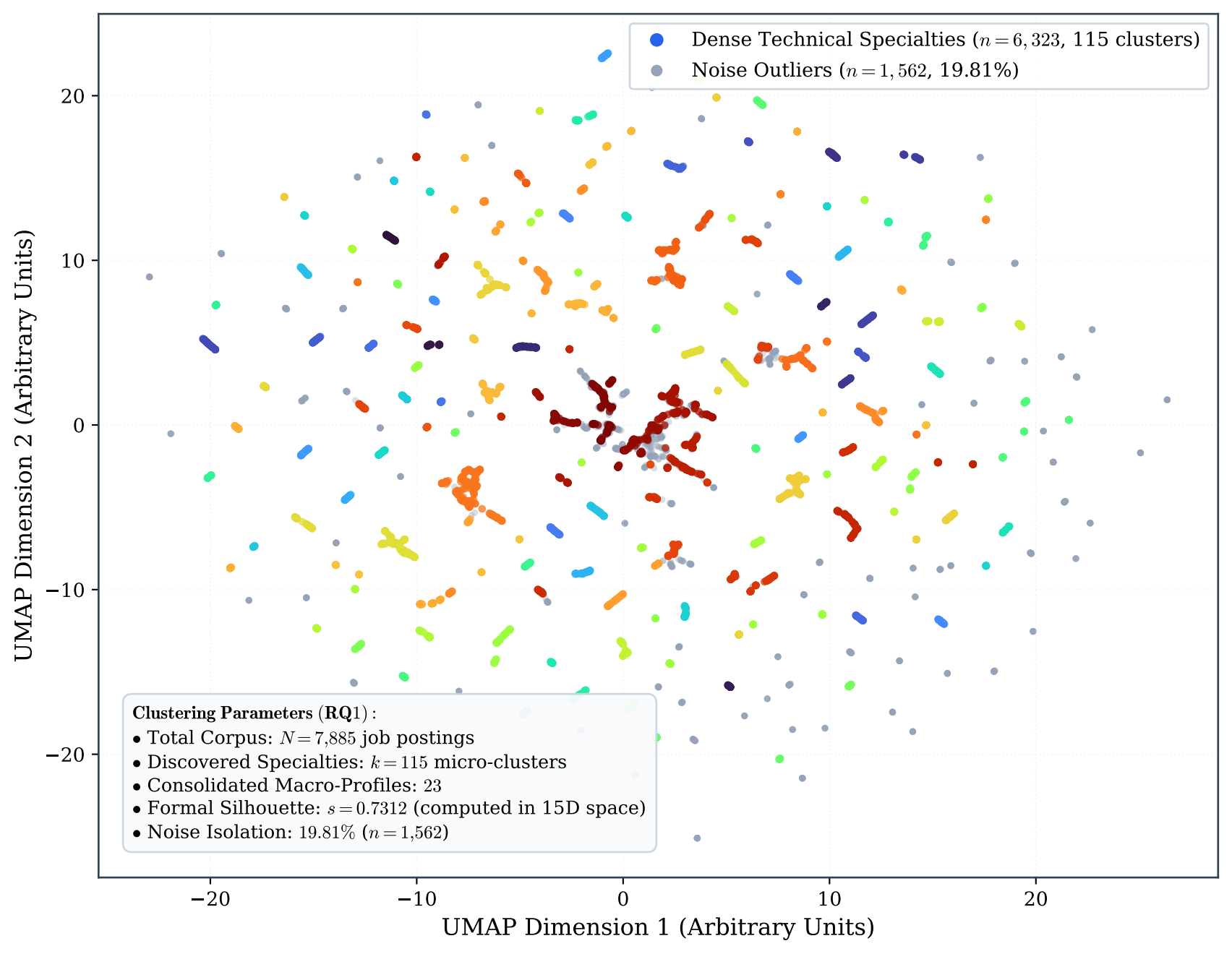

Dense Technical Specialties ($n = 6,323$, 115 clusters)
Noise Outliers ($n = 1,562$, 19.81%)
UMAP Dimension 2 (Arbitrary Units)
UMAP Dimension 1 (Arbitrary Units)
20
10
0
−10
−20
−20
−10
0
10
20
Clustering Parameters (RQ1):
• Total Corpus: $N = 7{,}885$ job postings
• Discovered Specialties: $k = 115$ micro-clusters
• Consolidated Macro-Profiles: 23
• Formal Silhouette: $s = 0.7312$ (computed in 15D space)
• Noise Isolation: 19.81% ($n = 1{,}562$)